Tugce Burcu Gundogdu

Computer Science Undergraduate Student
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RESEARCH INTERESTS

Information Inequality and Access, Human–Computer Interaction, Public-Interest Technology, Computational Social Science, AI-Mediated Information Environments.

EDUCATION

Bachelor of Science in Computer Science Expected May 2027
George Mason University, Fairfax, VA

PUBLICATIONS

Peer-Reviewed Journal Articles

1. Lynch, C. J., Jensen, E., Gore, R., Zamponi, V., O'Brien, K., Feldhaus, B., Smith, K., Martínez, J., Munro, M. H., Ozkose, T. E., **Gundogdu, T. B.**, Reinhold, A. M., Kavak, H., & Ezell, B. (2025). AI-generated messaging for life events using structured prompts: A comparative study of GPT with human experts and machine learning. *IEEE Access, 13*, 147002–147033. doi:10.1109/ACCESS.2025.3600146.

Conference Proceedings

2. **Gundogdu, T. B.**, Wang, J., Butler, B. S., & Lee, M. (2026). Local news at a distance: Psychological distance and the formation of news deserts. *Information Research, 31*(**iConf**), 1630–1638. doi:10.47989/ir31iConf64264.

3. **Gundogdu, T. B.**, Ozkose, T. E., Ozeren, F., Gore, R., Lynch, C. J., Zamponi, V., O'Brien, J., Ezell, B., & Kavak, H. (2025). Evolving risk communication: Analyzing risk response effectiveness of personalized narrative messages. In *2025 Systems and Information Engineering Design Symposium (SIEDS)* (pp. 92–97). IEEE. doi:10.1109/SIEDS65500.2025.11021204.

4. Wang, J., **Gundogdu, T. B.**, Ivic, R. K., Butler, B. S., & Lee, M. (2025). News deserts as information problems: A case study of local news coverage in Alabama. ***Proceedings of the Association for Information Science and Technology***, *62*(1), 741–753. doi:10.1002/pra2.1293.

5. Wang, J., **Gundogdu, T. B.**, Butler, B. S., & Lee, M. (2025). Spatial dynamics of local news: Mapping city co-mentions in Alabama. ***Proceedings of the Association for Information Science and Technology***, *62*(1), 1708–1710. doi:10.1002/pra2.1516. *Poster proceeding.*

RESEARCH EXPERIENCE

Community Informatics Lab, George Mason University Dec 2024 – Present
Undergraduate Research Assistant

- Investigate information inequality and news deserts through research on local news coverage; collected and analyzed 40,000+ news articles and contributed to peer-reviewed publications at ASIS&T and iConference.
- Conduct ongoing research on AI-mediated local news, examining how conversational interfaces may influence psychological distance, information access, and civic engagement.
- Contribute to SAFETI, a funded research project conducted in collaboration with the Virginia Department of Labor and Industry, examining workplace safety, workers' compensation policies, and regulatory impacts using large-scale administrative datasets.

Commonwealth Cyber Initiative (CCI), George Mason University 2025 – Present
CCI Scholar – Research Fellow

- Investigate methods for extracting and interpreting iOS geolocation data for digital forensic investigations, including stay-point detection and the identification of meaningful movement patterns.

Parkland Center for Clinical Innovation (PCCI) May – Aug 2026
Data Science Intern

- Independently led an advanced research project originally scoped for two interns, developing a Behavioral Lifestyle Index for Parkland Health covering approximately 500,000 patients and 9 million clinical encounters.
- Designed a reproducible methodology using large-scale data processing, feature selection, and clustering to reduce over 100 candidate variables to approximately 40 indicators; worked with clinicians to interpret results.
- Presented research to clinical and data-science audiences, including a session with 200+ attendees.

GMU Summer Team Impact Project (STIP) May – Nov 2025
Undergraduate Research Assistant

- Selected for an interdisciplinary undergraduate research program investigating human–wildlife interactions and potential zoonotic disease transmission risks.
- Collected and systematically coded social-media data from Facebook, TikTok, and Instagram to study human–deer interactions, contact behaviors, and associated public-health risks.
- Contributed to research presented at the 2025 American Public Health Association (APHA) Annual Meeting and co-authored five research posters.

Center for Social Complexity, George Mason University Dec 2023 – May 2025
Undergraduate Research Assistant

- Conducted research on LLM-generated and personalized risk communication, examining how different messaging approaches influence individuals' willingness to take protective actions during emergencies.
- Analyzed survey responses across five messaging frameworks and three risk domains, examining differences in message effectiveness and demographic response patterns.
- Led a team of three undergraduate researchers and collaborated with researchers at Old Dominion University, contributing to peer-reviewed publications.

RESEARCH GRANTS & FELLOWSHIPS

- **CCI Scholar Research Fellowship**, Commonwealth Cyber Initiative, 2025–2026 and 2026–2027.
- **Undergraduate Research Scholars Program (URSP) Research Grant**, George Mason University, Fall 2025. **$1,500** research project award; awarded following a student-written proposal.
- **Intensive Undergraduate Research Scholars Program (URSP)**, George Mason University, Summer 2026. **Selected for a $7,000 award; declined before participation** to pursue the PCCI research internship.
- **Summer Team Impact Project (STIP)**, George Mason University, Summer 2025. **$6,000** undergraduate research stipend.
- **Undergraduate Travel Fund**, Office of Outreach, Student Success & Engagement (OSSE), George Mason University, 2026. $350 travel support.

HONORS & AWARDS

- **First Place Best Poster Award**, OSCAR Poster Celebration, George Mason University, 2025.
- **Best Research Presentation Award**, OSCAR Research Presentation Competition, George Mason University, 2025. (News deserts research.)
- **Second Place Poster Award**, College of Public Health Celebration of Student Scholarship, George Mason University, 2026. (Co-authored poster.)
- **Third Place Poster Award**, AI-in-Gov Poster Competition, GMU AI-in-Gov Council, 2025. (Co-authored poster.)

ORAL RESEARCH PRESENTATIONS

1. **Gundogdu, T. B.**, Wang, J., Butler, B. S., & Lee, M. (2026). *Local news at a distance: Psychological distance and the formation of news deserts* [Paper presentation]. iConference 2026.

2. **Gundogdu, T. B.**, Ozkose, T. E., Ozeren, F., Gore, R., Lynch, C. J., Zamponi, V., O'Brien, J., Ezell, B., & Kavak, H. (2025). *Evolving risk communication: Analyzing risk response effectiveness of personalized narrative messages* [Paper presentation]. IEEE Systems and Information Engineering Design Symposium (SIEDS).

3. *News deserts research presentation* (2025). OSCAR Research Presentation Competition, George Mason University. **Best Research Presentation Award.**

Co-authored Conference Presentation

4. Roess, A., Von Hoene, E., Karably, B., Hopson, E., Ozkose, T. E., **Gundogdu, T. B.**, Ozeren, F., Hopgood, J., Ahmed, Z., & Anderson, T. (2025). *Characteristics of human, pet and deer contact captured on social media and implications for zoonotic disease transmission, 2015–2025* [Conference presentation]. *Flocking Together: Collaborations in One Health*, American Public Health Association (APHA) Annual Meeting & Expo, Washington, DC. **Co-author (not presenter).** Conference program.

POSTER PRESENTATIONS

2026

1. Karabiyik, U., Ozer, A. A., & **Gundogdu, T. B.** (2026). *From coordinates to context: Stay-point detection in iOS Cache.sqlite for digital forensics* [Poster presentation]. Commonwealth Cyber Initiative (CCI) Symposium. **Presenter.**

2. Lee, G., Hsu, J. H.-P., Desai, U. J., **Gundogdu, T. B.**, Lybarger, K., & Lee, M. (2026). *Advancing*

workplace safety through predictive analytics and policy evaluation [Poster presentation]. Innovation Week, George Mason University. **Presenter.**

3. Karably, B., Von Hoene, E., Hopson, E., Ozkose, T. E., **Gundogdu, T. B.**, Ozeren, F., Hopgood, J., Ahmed, Z., & Anderson, T. (2026). *Characteristics of human, pet and deer contact captured on social media and implications for zoonotic disease transmission, 2015–2025 – Methods* [Poster presentation]. College of Public Health Celebration of Student Scholarship, George Mason University. **Second Place Poster Award (co-author).**

2025

4. Wang, J., **Gundogdu, T. B.**, Butler, B. S., & Lee, M. (2025). *Spatial dynamics of local news: Mapping city co-mentions in Alabama* [Poster presentation]. ASIS&T Annual Meeting. **Presenter.**

5. **Gundogdu, T. B.**, Ozkose, T. E., Hopgood, J., Hopson, E., Ozeren, F., Ahmed, Z., Karably, B., Von Hoene, E., Anderson, T., & Roess, A. (2025). *An analysis of characteristics and behavior of humans and pets in deer interaction videos across Facebook, TikTok, Instagram* [Poster presentation]. OSCAR Poster Celebration, George Mason University. **Presenter; First Place Best Poster Award.**

6. Hopson, E., Ozkose, T. E., **Gundogdu, T. B.**, Ozeren, F., Hopgood, J., Ahmed, Z., Karably, B., Von Hoene, E., Anderson, T., & Roess, A. (2025). *A systematic approach for collecting and analyzing social media data for deer-human interactions on Facebook, TikTok and Instagram* [Poster presentation]. OSCAR Poster Celebration, George Mason University.

7. Ozkose, T. E., Hopson, E., **Gundogdu, T. B.**, Ozeren, F., Hopgood, J., Ahmed, Z., Karably, B., Von Hoene, E., Anderson, T., & Roess, A. (2025). *Deer and human contact as captured on Facebook, TikTok and Instagram: Showcasing viral videos* [Poster presentation]. OSCAR Poster Celebration, George Mason University.

8. Ozeren, F., **Gundogdu, T. B.**, Ozkose, T. E., Ahmed, Z., Hopgood, J., Hopson, E., Karably, B., Von Hoene, E., Roess, A., & Anderson, T. (2025). *Spatial analysis of deer and human interaction in geotagged videos on Facebook, TikTok, and Instagram* [Poster presentation]. OSCAR Poster Celebration, George Mason University.

9. Hsu, J. H.-P., **Gundogdu, T. B.**, Lybarger, K., & Lee, M. (2025). *An AI-based framework for understanding occupational injuries across Virginia* [Poster presentation]. AI-in-Gov Poster Competition, GMU AI-in-Gov Council. **Third Place Poster Award (co-author).**

2024

10. **Gundogdu, T. B.**, Wang, J., & Lee, M. (2024). *News deserts as information problems* [Poster presentation]. Innovation Week Poster Celebration, George Mason University. **Presenter.**

11. **Gundogdu, T. B.**, Ozkose, T. E., Ozeren, F., Gore, R., Lynch, C., & Kavak, H. (2024). *Using large language models for risk communication* [Poster presentation]. Undergraduate Research Colloquium, George Mason University. **Presenter.**

TEACHING EXPERIENCE

- **Undergraduate Teaching Assistant**, CS 112: Introduction to Computer Programming (Python), George Mason University, **Fall 2026**. Supported weekly labs and helped students debug Python programs.
- **Undergraduate Teaching Assistant**, CS 110: Essentials of Computer Science, George Mason University, **Fall 2024**. Graded assignments and provided foundational computing support.

WORKSHOPS & ACADEMIC ENGAGEMENT

- **Participant**, *Mapping Research and Practices on AI in the Public Sector*, Association for Information Science & Technology (ASIS&T) Annual Meeting, Arlington, VA, November 14, 2025.

RESEARCH & TECHNICAL SKILLS

- **Research Methods:** Survey design and analysis, literature reviews, research synthesis, qualitative coding, policy evaluation, interdisciplinary study design.
- **Quantitative and Computational Methods:** NLP and LLM-assisted text analysis, geographic and social-network analysis, clustering, feature selection, predictive modeling, descriptive statistics, data visualization.
- **Programming and Data:** Python (pandas, NumPy, Matplotlib), R, SQL, Spark, Java, C.
- **Research Tools:** Qualtrics, Stata, Tableau, Git/GitHub, Jupyter, Linux/Unix.
- **Languages:** English (native), Turkish (native), French (intermediate).

PROFESSIONAL & CAMPUS AFFILIATIONS

- **Professional Memberships:** Association for Information Science & Technology (ASIS&T); American Public Health Association (APHA), student member.
- **University Outreach:** Student Assistant, GMU Office of Outreach, Student Success & Engagement, 2023–2025.
- **George Mason University Clubs:** Digital Law Club; Model United Nations Club; AI Club; Book Buzz Club.